

**Electronic and Telecommunication Engineering
University of Moratuwa, Sri Lanka**



EN4730 - Convex Engineering Design

Assignment 02

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EN4730 Convex Engineering Design Assignment 02

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Objective

In this assignment, you will explore the connection between saddle points of the Lagrangian and the concept of strong duality, using a game-theoretic perspective.

1. Follow the steps below to analyze and interpret the mini-max inequality and its implications in the context of optimization and zero-sum games.

Understand the game setup.

- Suppose two teams are involved in a cricket match with payoff function $f(w, z)$, where $f(w, z)$ represents the total runs scored by Team B in an over.
- Team A (bowling) chooses $w \in W$ to *minimize* runs.
- Team B (batting) chooses $z \in Z$ to *maximize* runs.
- For simplicity, we approximate this over-level interaction as a zero-sum game: Team B's gain in runs corresponds directly to Team A's loss.

(a) What is the meaning of $\inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$? [05 marks]

(b) What is the meaning of $\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right)$? [05 marks]

Analyze the order of play.

- Consider two scenarios: (a) Team A moves first, and (b) Team B moves first.
(a) Use reasoning to explain why the inequality

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$$

always holds.

[20 marks]

①

- a) minimum number of runs Team A can guarantee, assuming Team B responds optimally.
- b) maximum number of runs Team B can guarantee, even against best bowling response.

Case 1 team A moves first

- A chooses w
- B responds optimally

$$\inf_w \left(\sup_z f(w, z) \right)$$

- A is disadvantaged because B reacts after seeing w .

Case 2 team B moves first

- B chooses z
- A responds optimally

$$\sup_z \left(\inf_w f(w, z) \right)$$

- B is disadvantaged because A reacts after seeing z .

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right) \text{ inequality always holds}$$

proof for any fixed $w \in W$ and $z \in Z$,

$$\inf_{w' \in W} f(w', z) \leq f(w, z) \leq \sup_{z' \in Z} f(w, z')$$

greatest lower bound $\leq$ function $\leq$ least upper bound

for every w, z

$$\inf_{w \in W} f(w, z) \leq \sup_{z \in Z} f(w, z)$$

Take supremum over z on LHS,

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \sup_{z \in Z} f(w, z) \quad \forall w \in W$$

Take infimum over w on RHS,

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$$

$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right)$ - Team B chooses first, then Team A minimizes.
Team B gets a lower guaranteed payoff.

$\inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$ - Team A chooses first, then Team B maximizes.
Team B can get a higher payoff.

so

$$\sup_{z \in Z} \left(\inf_{w \in W} f(w, z) \right) \leq \inf_{w \in W} \left(\sup_{z \in Z} f(w, z) \right)$$

- (b) Consider a single over. Suppose Team A has two bowling strategies: Defensive(w_1) and Aggressive(w_2), and Team B has two batting strategies: Conservative(z_1) and Aggressive(z_2). The payoff table (runs scored by Team B in an over) is:

	z_1 (Conservative)	z_2 (Aggressive)
w_1 (Defensive)	4	10
w_2 (Aggressive)	2	12

i. **Minimax:**

If Team A bowls first, compute $\inf_w \left(\sup_z f(w, z) \right)$ and explain its meaning in the context of the over. **[15 marks]**

ii. **Maximin:**

If Team B bats first, compute $\sup_z \left(\inf_w f(w, z) \right)$ and explain its meaning in the context of the over. **[15 marks]**

iii. **Saddle Point:**

Verify the minimax inequality $\sup_z \left(\inf_w f(w, z) \right) \leq \inf_w \left(\sup_z f(w, z) \right)$ and determine if a saddle point exists. **[25 marks]**

$$i) \quad \text{for } w_1, \quad \sup_z f(w, z) = \max(4, 10) = 10$$

$$\text{for } w_2, \quad \sup_z f(w, z) = \max(2, 12) = 12$$

$$\inf_w \left(\sup_z f(w, z) \right) = \min(10, 12) = 10$$

If team A bowls first, it will choose w_1 (defensive), ensuring that even if Team B responds optimally, the maximum runs conceded is 10.

$$\text{ii) for } z_1, \quad \inf_w f(w, z) = \min(2, 4) = 2$$

$$\text{for } z_2, \quad \inf_w f(w, z) = \min(10, 12) = 10$$

$$\sup_z \left(\inf_w f(w, z) \right) = \max(2, 10) = 10$$

If team B bats first, it will choose z_2 (aggressive), ensuring that even if Team A responds optimally, the minimum runs scored is 10.

$$\text{ii) } \sup_z \left(\inf_w f(w, z) \right) = 10$$

$$\inf_w \left(\sup_z f(w, z) \right) = 10$$

$$\sup_z \left(\inf_w f(w, z) \right) = \inf_w \left(\sup_z f(w, z) \right)$$

$$\therefore \sup_z \left(\inf_w f(w, z) \right) \leq \inf_w \left(\sup_z f(w, z) \right) \text{ also holds.}$$

$$\text{Since } \sup_z \left(\inf_w f(w, z) \right) = \inf_w \left(\sup_z f(w, z) \right)$$

Saddle point exists

saddle point occurs at $(w_1, z_2) = 10$ runs.

2. Write-down the equivalent linear programming (LP) of the following optimization problems.

(a)

$$\begin{aligned} &\text{minimize } \|x\|_1, \\ &\text{subject to } \|Ax - b\|_\infty \leq 1. \end{aligned}$$

$$\text{LP format} \Rightarrow \min C^T y \quad \text{s.t.} \quad Gy \leq h$$

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

introduce auxiliary variables $t_i \geq 0$ s.t.

$$|x_i| \leq t_i \Leftrightarrow -t_i \leq x_i \leq t_i$$

$$x - t \leq 0 \quad - \textcircled{1} \quad -x - t \leq 0 \quad - \textcircled{2}$$

$$\text{objective} \quad \min \sum_{i=1}^n t_i$$

$$\|Ax - b\|_\infty \leq 1 \Leftrightarrow -1 \leq (Ax - b)_j \leq 1 \quad \forall j$$

$$Ax - b \leq 1 \quad - (Ax - b) \leq 1$$

$$Ax \leq b + 1 \quad - \textcircled{3} \quad -Ax \leq 1 - b \quad - \textcircled{4}$$

$$t \geq 0 \Rightarrow -t \leq 0 \quad - \textcircled{5}$$

Equivalent LP formulation

$$\text{minimize } \sum_{i=1}^n t_i$$

$$\text{subject to } x - t \leq 0$$

$$-x - t \leq 0$$

$$Ax \leq b + 1$$

$$-Ax \leq 1 - b$$

$$-t \leq 0$$

2. Write-down the equivalent linear programming (LP) of the following optimization problems.

(b)

$$\text{minimize } \|Ax - b\|_1.$$

$$\|Ax - b\|_1 = \sum_{j=1}^m |(Ax - b)_j|$$

introduce $u_j \geq 0$ s.t.

$$|(Ax - b)_j| \leq u_j \Leftrightarrow -u_j \leq (Ax - b)_j \leq u_j$$

$$Ax - b \leq u$$

$$-(Ax - b) \leq u$$

$$Ax - u \leq b \quad \text{--- ①}$$

$$-Ax - u \leq -b \quad \text{--- ②}$$

$$u \geq 0 \quad \text{--- ③}$$

Equivalent LP formulation

$$\text{minimize} \quad \sum_{j=1}^m u_j$$

$$\text{subject to} \quad Ax - u \leq b$$

$$-Ax - u \leq -b$$

$$-u \leq 0$$